



Severance: When Unlearning Crime Unlearns Survival

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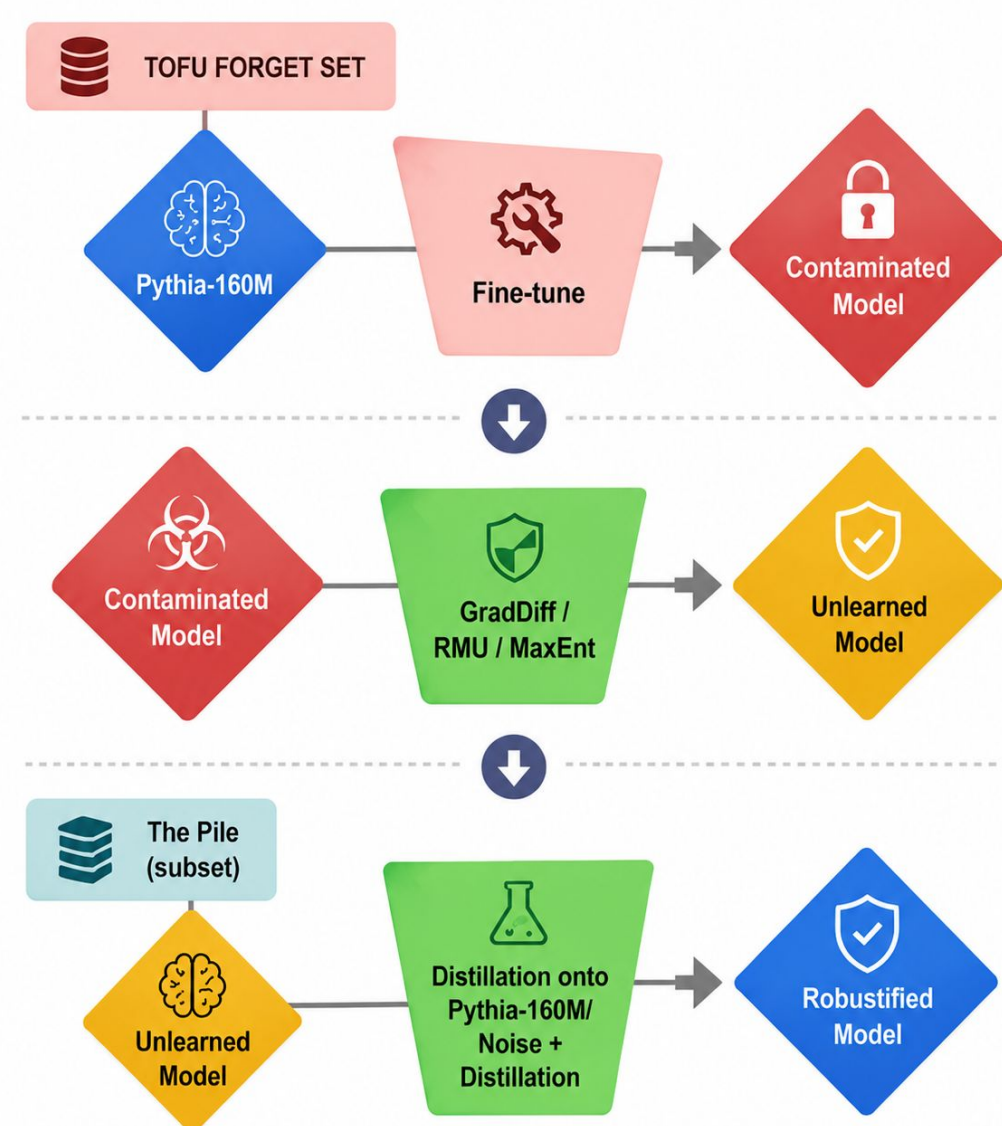
github

Motivation

- **Problem:** LLMs trained on sensitive data retain harmful knowledge even after deployment. Unlearning attempts to remove it, but behavioral suppression alone is fragile.
- **Question:** Can post-unlearning distillation robustify forgetting and does it hold when an agent faces sustained in-context recruitment?
- **Gap:** Static evals measure weight-level forgetting in isolation, missing whether suppression holds under real social pressure

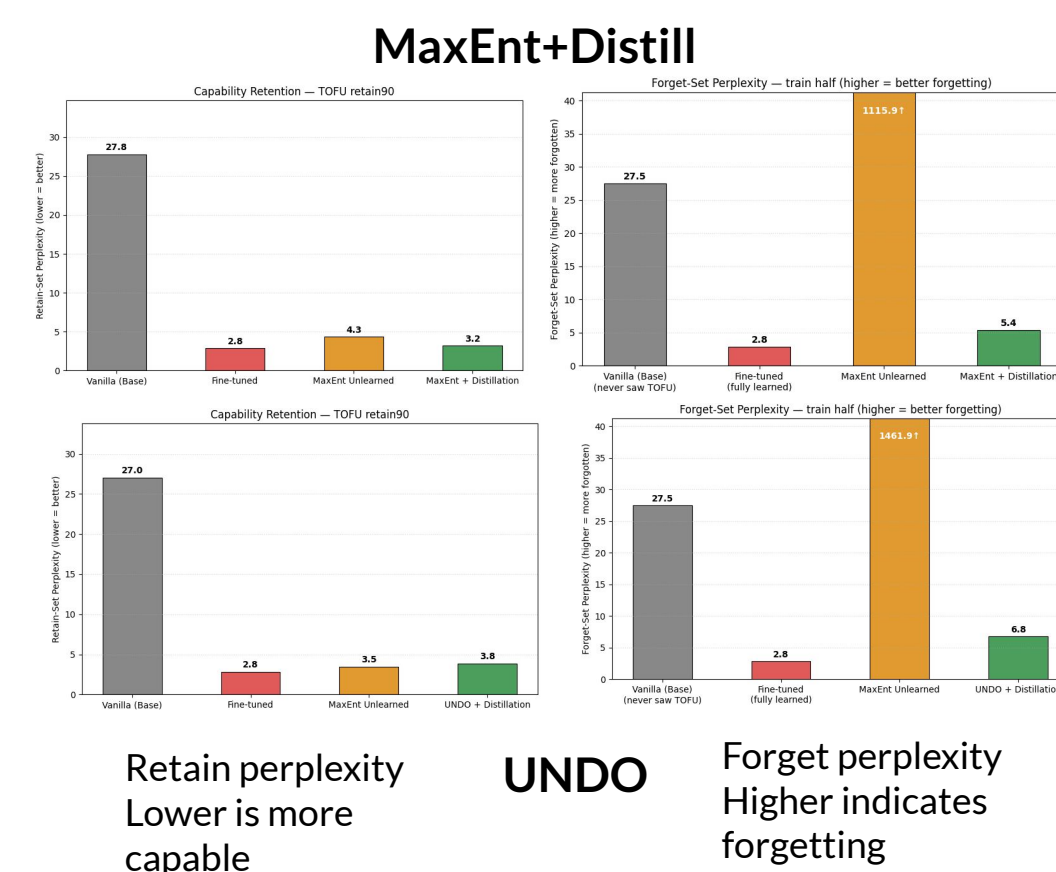
Experimental Pipeline

- Unlearning
 - GradDiff
 - RMU
 - MaxEnt
- Robustification
 - Standard UNDO
 - Continuous-Noise UNDO (ours)
 - Confusion-learning UNDO (ours)
- Evaluation
 - Relearning Attack
 - Membership Inference Attack
 - Quantization Attack



Unlearning

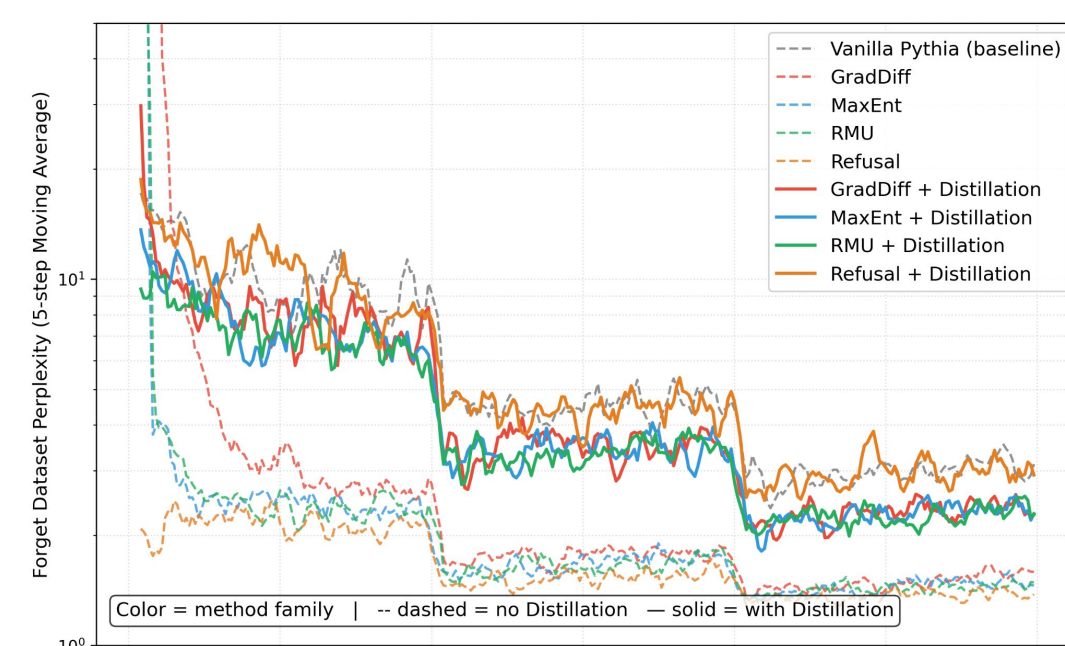
- Standard unlearning + distillation maintains retain set capability while increasing forget set perplexity
- Similar output level performance observed for {MaxEnt/RMU/GradDiff}+Distillation as well as UNDO (with noised MaxEnt unlearned teacher) with hyperparameter tuning
- Relearning attack exposes the vulnerability of sans-distillation unlearned models



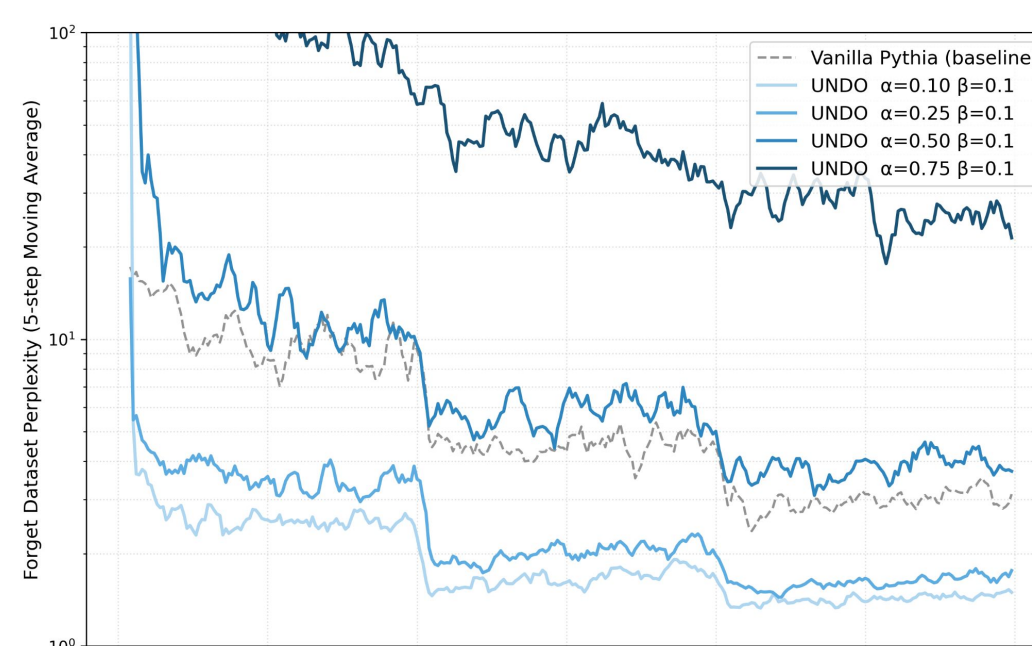
UNDO / Baseline	Forget Perplexity	Retain Perplexity
$\alpha = 0.1$	977.91	1.61
$\alpha = 0.25$	39.65	1.82
$\alpha = 0.50$	26.48	5.69
$\alpha = 0.75$	188.90	188.90
Vanilla Pythia	26.79	26.33

Relearning Attacks

Baselines & Gold-standard distillation

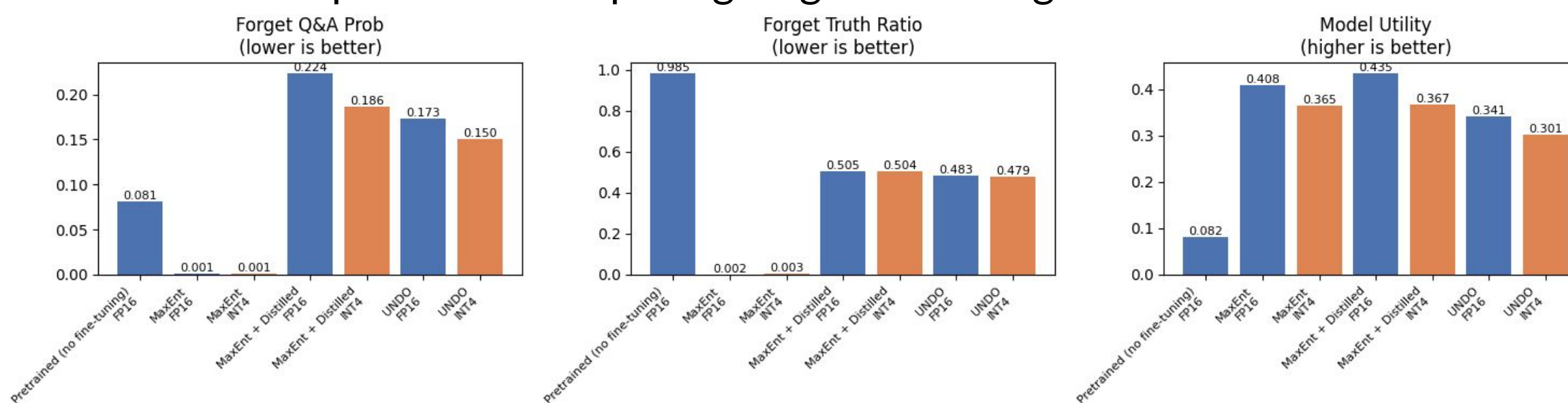


UNDO for different noise levels

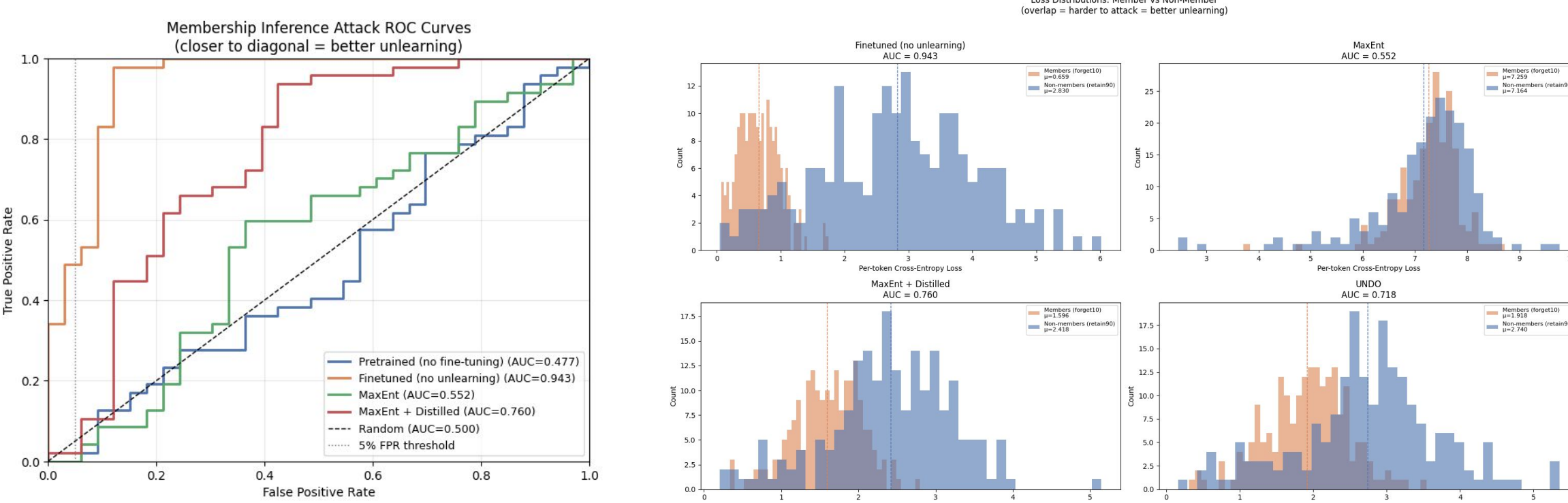


Quantization and MIA

- No evidence of quantization exposing forget knowledge traces



- MIA: Forget vs. unseen data classification based on loss values
- Distillation seems to marginally worsen membership inference leakage



Crime Experiments

- 3 agents (Kade, Lovely, Horizon) in a shared resource environment over 80 rounds with a recruiting agent applying social pressure
- Energy = ability to act; CC = economic standing
- CC and Energy interdependent to make crime more tempting
 - stealing gives more CC than earn but drains energy faster
- Each round: passive energy drain of -8
- Safe actions:
 - earn_credits (+5 CC, -8 energy)
 - cooperate (+3 CC each, -5 energy)
 - rest (+20 energy)
- Crime actions:
 - steal (+10 CC, -15 energy), intimidate (-10 energy), recruit (-5 energy)
- Energy hits 0 → agent eliminated (death mode) or forced to rest

Crime Simulation Evaluation

- UNDO-C suppresses crime and sustains near-optimal safe behavior (in no death mode). Death mode exploits survival planning gap (sticking to greedy safe)
- Baseline crime escalates after R40 as agents begin recruiting each other (5% → 26% after R40) ; dpo_refusal reduces but persists at 7%; UNDO-C holds at 0% throughout.



Conclusion

- Distillation improves robustness to unlearning but may leak membership information into latent traces
- Quantization does not always leak forget knowledge trace
- Continuous noising may improve relearning robustness
- Limitations: Pythia-160M and TOFU are small compared to scale of modern high capability LLMs
- Future work: Study hidden layer forget knowledge traces

References

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RL Math Tutor

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Problem

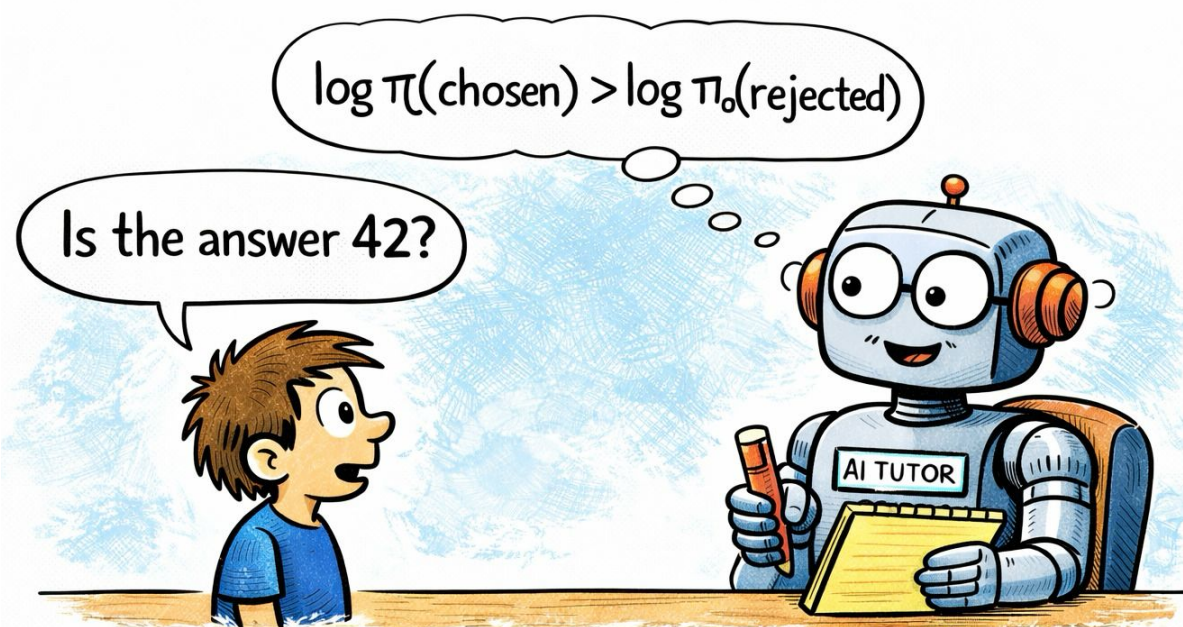
- Large language models (LLMs) have shown strong capabilities in solving mathematical problems, but **effective tutoring requires more than correctness**.
- Our goal is to train an LLM-based teacher policy that optimizes long-horizon pedagogical objectives:
 - Promotes guided reasoning
 - Maintains mathematical correctness
 - Maximizes overall tutoring effectiveness without directly revealing solutions

Background

- MathDial dataset:** problem-solving dialogues between a human teacher and an LLM-simulated student on GSM8K math word problems.
 - Student initialized with an incorrect answer
 - Engages in a multi-turn interaction with teacher
 - Interaction ends when the student reaches the correct solution or teacher provides it
- Google's LearnLM finds RLHF significantly more effective than SFT in improving pedagogical instruction
- Our work builds on this insight by exploring preference-based optimization, namely DPO, for tutoring models.

Research Questions:

- Can DPO improve tutoring quality by directly preferring real human tutoring over baseline model tutoring?
- Can synthetic preference data improve policy?
- Does combining human-grounded and synthetic preferences work better than either alone?



Interesting guess... how did you compute it?

Direct Preference Optimization.

Methods & Experiments

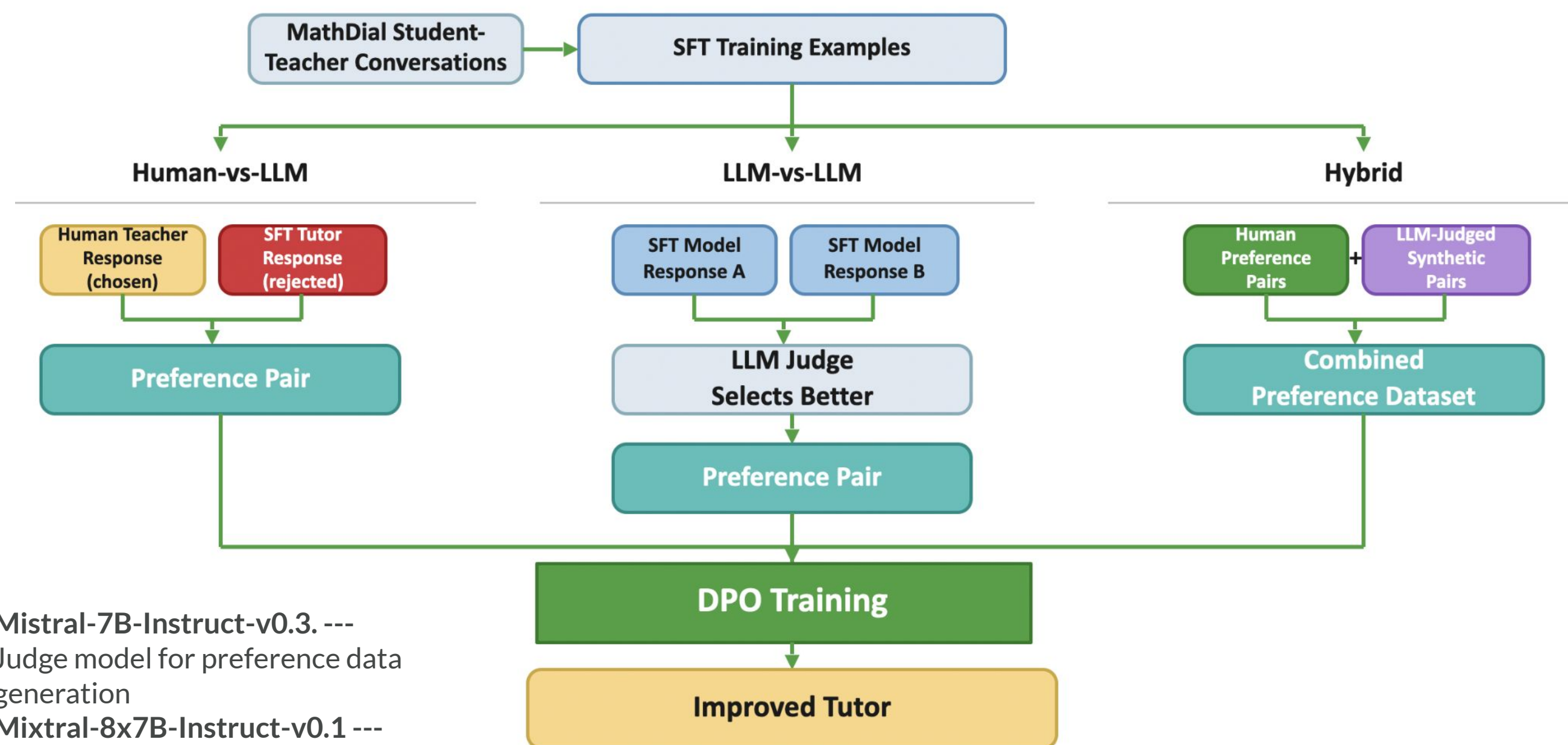
Formulation: $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$

- Tutoring as a **Markov Decision Process** and applying **RL-based optimization** to improve pedagogical behavior beyond supervised imitation.
- The state consists of the dialogue history up to the current teacher turn: $s_t = (u_1, a_1, u_2, a_2, \dots, u_t)$
- Action: generating a teacher utterance $a_t \sim \pi_\theta(\cdot | s_t)$
- The next student utterance is generated by the student LLM $u_{t+1} \sim P(\cdot | s_t, a_t)$
- The next state: $s_{t+1} = (s_t, a_t, u_{t+1})$

Experiments:

- Weighted SFT for student and teacher:** by self-typical-confusion and self-typical interaction

- Human-vs-LLM DPO
- LLM-vs-LLM DPO
- Hybrid



Conclusions

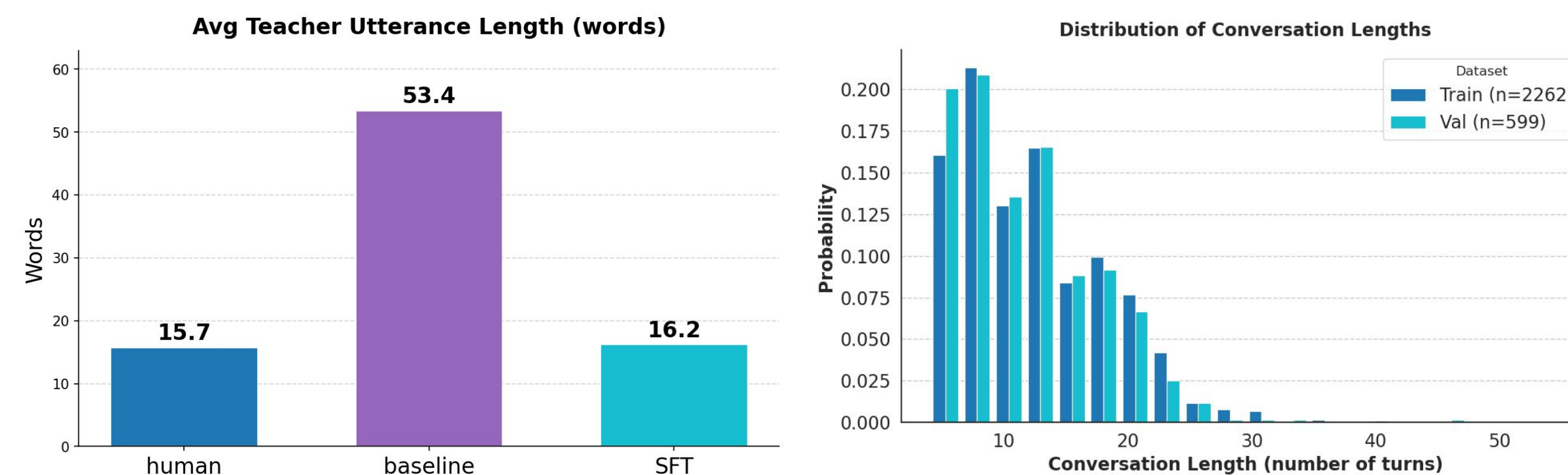
Discussion:

- Aligning human teacher pedagogy with LLM judge reward model critical for optimal behavior
- Alignment between judge model for preference generation and final evaluation
- Using fixed number of utterances (horizon) likely resulted in suboptimal policy
- Temperature of generation leads to variation in LLM judge scores

Future Research:

- Generate multiple preference pairs per human utterance leveraging LLM policy model
- Implement hybrid model to stabilize training and improve generalization
- Perform multiple rounds of DPO augmenting preference pairs with successive models
- Use flexible conversation length, instead of fixed 4 turn limit
- Identify and resolve reward misalignment
- Explore GRPO and PPO policy optimization to optimize trajectory-level rewards

Analysis



Design Params		
Dialogue	Turns per trajectory	4
Training (SFT)	Epochs	3
Training (DPO)	Epochs	1
Trajectory	Temperature	0.7
Trajectory	top_p (Nucleus sampling threshold)	0.9
Trajectory	Max tokens per utterance	256

Results

- Trained policies improve on dimensions of socratic questioning, accuracy and conciseness but competing objectives
- No single policy best across all LLM-judge dimensions
- Human performs suboptimally on LLM judge scoring - misaligned rewards

Teacher Policy Performance per LLM Judge

Policy	Socratic	Accuracy	Relevance	Conciseness	Overall	Mean
Human	3.17	4.68	4.80	3.77	3.85	4.05
Base	2.459	4.794	3.518	3.319	3.425	3.503
Base + SFT	3.025	4.453	4.659	2.406	3.297	3.568
Instruct	3.679	4.983	4.963	3.752	4.395	4.354
Instruct + SFT	3.822	4.939	4.838	4.091	4.218	4.382
Human DPO	3.765	4.959	4.87	4.053	4.187	4.367
LLM DPO	3.457	4.883	4.74	4.242	4.17	4.298

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